Recap: Fourier series

- If $f(x)$ is piecewise smooth on $(-L, L]$ then its Fourier series is

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(k_n x) + b_n \sin(k_n x), \qquad k_n = \frac{n\pi}{L}$$

  where $f(x)$ is continuous

- The coefficients are given by Euler-Fourier formulae
- We denote the periodic extension of $f(x)$ as $F(x)$
- If $F(x)$ continuous then Fourier series convergence is uniform (e.g., $f = x^2$)
- If $F(x)$ is discontinuous then Gibbs phenomenon occurs (e.g., $f = x$)

Lecture 18

In Lecture 18, we will:

1. Define complex Fourier series
2. Define even/odd periodic extensions of functions and calculate their Fourier series

Complex exponentials create a more compact Fourier series

$e^{ix} = \cos x + i \sin x$

Definition

- The function $f(x)$ on $(-L, L]$ has the *complex Fourier series*

$$\sum_{n=-\infty}^{n=\infty} c_n \mathrm{e}^{\mathrm{i}k_n x} \quad \text{where} \quad c_n = \frac{1}{2L}\int_{-L}^{L} f(x)\mathrm{e}^{-\mathrm{i}k_n x}\mathrm{d}x$$

$$\int_{-L}^{L} e^{ik_n x} e^{-ik_m x}\,dx = \frac{1}{i(k_n-k_m)}\underbrace{\left[e^{i(k_n-k_m)x}\right]_{-L}^{L}}_{e^{i(n-m)\pi} - e^{-i(n-m)\pi}}$$

$$= 2i\sin((n-m)\pi) = 0$$

for $n \neq m$

$k_n - k_m$
$= \frac{n\pi}{L} - \frac{m\pi}{L}$
$= (n-m)\pi/L$
$\frac{1}{2i}(e^{ix} - e^{-ix})$
$= \sin x$

$$\int_{-L}^{L} e^{ik_n x} e^{-ik_n x}\,dx = 2L$$

$$\int_{-L}^{L} f(x) e^{-ik_m x}\,dx = \int_{-L}^{L}\sum_{n=-\infty}^{\infty} c_n e^{ik_n x} e^{-ik_m x}\,dx$$

$$= 2L c_m$$

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- This will be our starting point for Fourier transforms (Topic 6)

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Exercise 4.6: Calculate the complex Fourier series of $f(x) = x$ on $(-\pi, \pi]$

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- This will be our starting point for Fourier transforms (Topic 6)

Exercise 4.6: Calculate the complex Fourier series of $f(x) = x$ on $(-\pi, \pi]$

Recall that for $f(x) = x$ with $L = \pi$ and $k_n = n$

$$F(x) = -2\sum_{n=1}^{\infty} \frac{(-1)^n}{n}\sin(nx)$$

where it is continuous.

For $n \neq 0$

$$2\pi\, c_n = \int_{-\pi}^{\pi} x\, \mathrm{e}^{-\mathrm{i}nx}\, \mathrm{d}x = \left[\frac{x\, \mathrm{e}^{-\mathrm{i}nx}}{-\mathrm{i}n}\right]_{-\pi}^{\pi} + \frac{1}{\mathrm{i}n} \int_{-\pi}^{\pi} \mathrm{e}^{-\mathrm{i}nx}\, \mathrm{d}x$$

$$= \frac{2\pi(-1)^n}{-\mathrm{i}n} + \left[\frac{\mathrm{e}^{-\mathrm{i}nx}}{n^2}\right]_{-\pi}^{\pi} = \frac{2\pi\mathrm{i}(-1)^n}{n}$$

For $n \neq 0$

$$\cancel{2\pi}\, c_n = \int_{-\pi}^{\pi} x\, \mathrm{e}^{-\mathrm{i}nx}\, \mathrm{d}x = \left[\frac{x\, \mathrm{e}^{-\mathrm{i}nx}}{-\mathrm{i}n}\right]_{-\pi}^{\pi} + \frac{1}{\mathrm{i}n}\int_{-\pi}^{\pi} \mathrm{e}^{-\mathrm{i}nx}\, \mathrm{d}x$$

$$= \frac{2\pi(-1)^n}{-\mathrm{i}n} + \left[\frac{\mathrm{e}^{-\mathrm{i}nx}}{n^2}\right]_{-\pi}^{\pi} = \frac{\cancel{2\pi}\mathrm{i}(-1)^n}{n}$$

$$\text{and} \qquad 2\pi\, c_0 = \int_{-\pi}^{\pi} x\, \mathrm{d}x = 0$$

$$F(x) = \sum_{\substack{n=-\infty \\ \neq 0}}^{\infty} \frac{i(-1)^n}{n} e^{inx} = \sum_{n=1}^{\infty} \frac{\cancel{i}(-1)^n}{n} \underbrace{\left(e^{inx} - e^{-inx}\right)}_{\cancel{-2i}\sin(nx)}$$

$$= -2\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin(nx)$$

Fourier series of even and odd functions

Consider the Fourier series of $f(x)$ on $(-L, L]$, i.e.,

$$\frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(k_n x) + b_n \sin(k_n x) \qquad \text{where} \qquad k_n = \frac{n\pi}{L}$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos(k_n x)\,\mathrm{d}x \quad \text{and} \quad b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin(k_n x)\,\mathrm{d}x$$

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0

Case f even, i.e., $f(-x) = f(x)$

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Case f even, i.e., $f(-x) = f(x)$

The Fourier series simplifies to

$$\frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(k_n x) \quad \text{where} \quad a_n = \frac{2}{L}\int_{0}^{L} f(x)\cos(k_n x)\mathrm{d}x$$

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Fourier series of even and odd functions

Consider the Fourier series of $f(x)$ on $(-L, L]$, i.e.,

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$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos(k_n x)\,\mathrm{d}x \quad \text{and} \quad b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin(k_n x)\,\mathrm{d}x$$

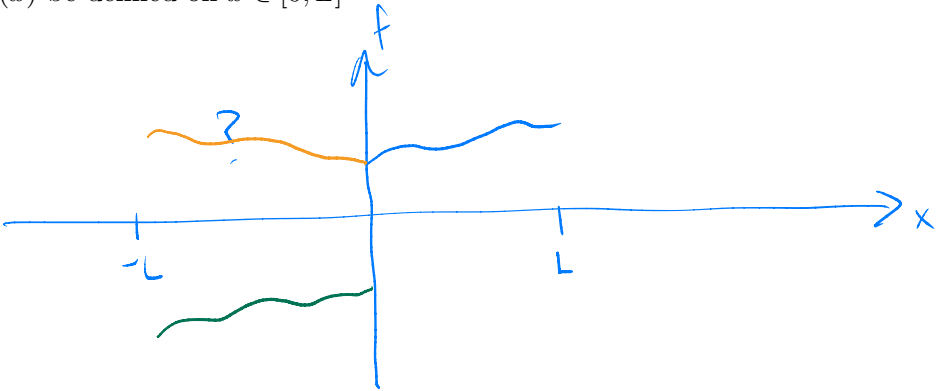
Case f odd, i.e., $f(-x) = -f(x)$

The Fourier series simplifies to

$$\sum_{n=1}^{\infty} b_n \sin(k_n x) \quad \text{where} \quad b_n = \frac{2}{L}\int_{0}^{L} f(x)\sin(k_n x)\mathrm{d}x$$

Even and odd extensions

- Let $f(x)$ be defined on $x \in [0, L]$



Even and odd extensions

- Let $f(x)$ be defined on $x \in [0, L]$
- We can calculate its *even periodic extension* $F(x)$, such that

$$
\begin{aligned}
F(x) &= f(x) \quad \text{for} \quad x \in [0, L] \\
F(x) &= f(-x) \quad \text{for} \quad x \in (-L, 0) \\
F(x + 2L) &= F(x)
\end{aligned}
$$

Even and odd extensions

- Let $f(x)$ be defined on $x \in [0, L]$
- We can calculate its *even periodic extension* $F(x)$, such that

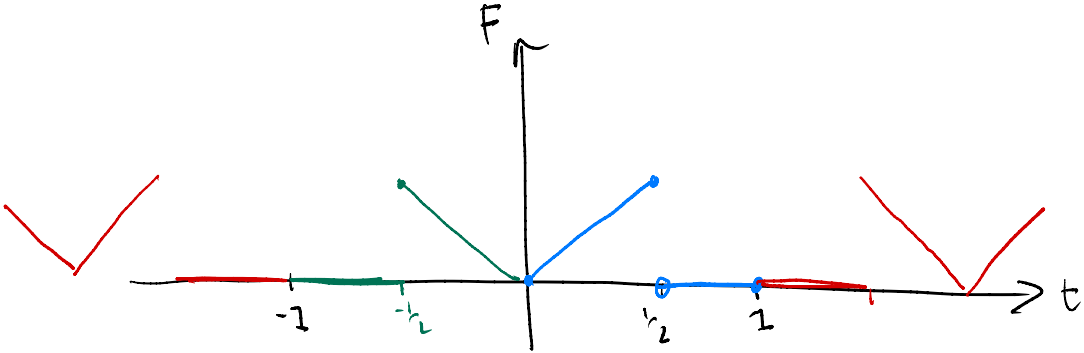
$$\begin{aligned} F(x) &= f(x) \quad \text{for} \quad x \in [0, L] \\ F(x) &= f(-x) \quad \text{for} \quad x \in (-L, 0) \\ F(x + 2L) &= F(x) \end{aligned}$$

- We can also calculate its *odd periodic extension* $F(x)$, such that

$$\begin{aligned} F(x) &= f(x) \quad \text{for} \quad x \in [0, L] \\ F(x) &= -f(-x) \quad \text{for} \quad x \in (-L, 0) \\ F(x + 2L) &= F(x) \end{aligned}$$

Exercise 4.7: Find the Fourier series of the even periodic extension of

$$f(t) = \begin{cases} t, & 0 \leqslant t \leqslant 1/2 \\ 0, & 1/2 < t \leqslant 1 \end{cases}$$



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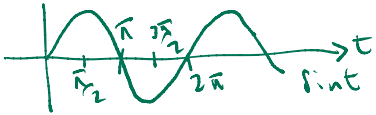


$$f(t) = \begin{cases} t, & 0 \leqslant t \leqslant 1/2 \\ 0, & 1/2 < t \leqslant 1 \end{cases}$$

$L = 1$

$k_n = \frac{n\pi}{L} = n\pi$

$$a_0 = 2\int_0^1 f(t)\,dt = 2\int_0^{1/2} t\,dt = 2\left[\tfrac{1}{2}t^2\right]_0^{1/2} = 2\cdot\tfrac{1}{2}\left(\tfrac{1}{2}\right)^2 = \tfrac{1}{4}$$

$$a_n = 2\int_0^{1/2} t\cos(n\pi t)\,dt = 2\left\{\left[\frac{t}{n\pi}\sin(n\pi t)\right]_0^{1/2} - \frac{1}{n\pi}\int_0^{1/2}\sin(n\pi t)\,dt\right\}$$

$$= \cancel{2}\left\{\frac{1}{\cancel{2}n\pi}\sin\left(\frac{n\pi}{2}\right) + \frac{2}{(n\pi)^2}\left[\cos(n\pi t)\right]_0^{1/2}\right\}$$

$$= \frac{1}{n\pi}\sin\left(\frac{n\pi}{2}\right) + \frac{2}{(n\pi)^2}\left[\underbrace{\cos\left(\frac{n\pi}{2}\right)}_{=(-1)^m} - 1\right]$$

$= 0 \quad n = 2m$

$= -(-1)^m \quad n = 2m-1$

$n = 2m$; 0 $\quad n = 2m-1$

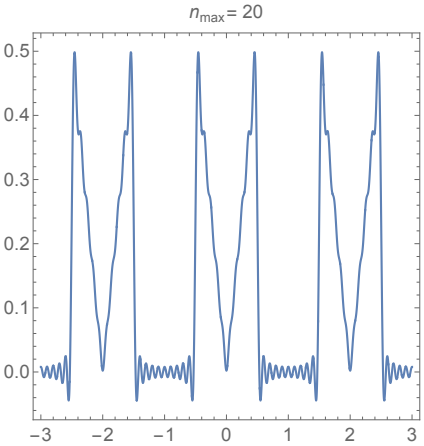
$$F(t) = \frac{1}{2}\cdot\frac{1}{4} + \sum_{m=1}^{\infty} \frac{2}{(2m)^2\pi^2}\left((-1)^m - 1\right)\cos(2m\pi t)$$

$$- \sum_{m=1}^{\infty}\left(\frac{(-1)^m}{(2m-1)\pi} + \frac{2}{(2m-1)^2\pi^2}\right)\cos((2m-1)\pi t)$$

Nb. $(-1)^m - 1 = \begin{cases} 0 & \text{if } m = 2l \\ -2 & \text{if } m = 2l-1 \end{cases}$

$$= \frac{1}{8} - \sum_{l=1}^{\infty} \frac{4}{(4l-2)^2\pi^2}\cos((4l-2)\pi t)$$

$$- \sum_{m=1}^{\infty} \cdots \quad \text{where cts}$$


$n_{max} = 20$
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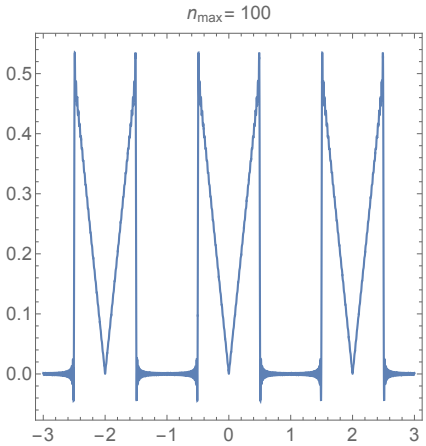
0.4
0.3
0.2
0.1
0.0
−3
−2
−1
0
1
2
3


$n_{max} = 100$
0.5
0.4
0.3
0.2
0.1
0.0
−3
−2
−1
0
1
2
3